Simulation Report: BJT/MOSFET Characteristics and Differential Amplifier Performance

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Fundamentals of Analogue Circuits — Experiment 1 (LTspice)

Abstract. This report summarizes LTspice simulations for transistor characteristics and a MOS differential amplifier. The BJT test used a 2N2222 transistor with swept collector voltage and base current. The differential-amplifier section used a matched 2N7002 pair, 10 kΩ drain resistors, ±12 V supplies, and a 2.577 mA tail current. The recorded transient plots show a large differential output for differential input and near-zero differential output for common-mode input.

Scope. The discussion is based on the uploaded laboratory guide and the screenshots in the submitted DOCX file. Numeric values are estimated from the visible plots, because the screenshots do not include LTspice cursor readouts or raw data tables.

Item	Main setting / observation
BJT characteristic test	2N2222; V1 swept 0–15 V in 10 mV steps; I1 swept 0–100 µA in 10 µA steps. Collector-current family indicates active and saturation regions.
NMOS characteristic setup	NMOS device connected with independent gate-source and drain-source voltage sources. The uploaded screenshot shows the circuit setup; no separate MOSFET output plot was available.
Differential amplifier	Two 2N7002 MOSFETs, RD1 = RD2 = 10 kΩ, RG1 = RG2 = 10 kΩ, ±12 V supplies, I_tail = 2.577 mA.
Measured gain estimate	Differential output ≈ ±1.2 V for 10 mV input amplitude, so A_vd ≈ 120 V/V (≈ 40.8 dB).
Common-mode result	For 100 mV common-mode input, the differential output is visually near 0 mV; using <1 mV as an upper bound gives A_cm < 0.01 and CMRR > 1.2×10^4 (≈72 dB).

1. Objectives and Theory

The experiment has two objectives: first, to obtain current-voltage characteristics of transistor devices using LTspice; second, to build a differential amplifier and evaluate differential gain, common-mode gain, and common-mode rejection ratio.

For a bipolar junction transistor in the forward-active region, the collector current is approximately exponential with base-emitter voltage:

$$\mathbf{IC = IS\ exp(VBE/VT), \quad IB \approx IC/\beta, \quad VT \approx 25\ mV}$$

When VCE is very small, the BJT enters saturation and IC is strongly affected by VCE. When VCE is sufficiently high, the transistor moves into the active region and IC is mainly controlled by base current. A small positive slope in the active region is expected because of channel-width modulation/Early effect.

For an NMOS transistor, the drain current is controlled by the gate-source voltage VGS and the drain-source voltage VDS. Below threshold, the current is small. Above threshold, the output curves rise with VDS in the triode region and become flatter in saturation.

For the differential amplifier, the differential output voltage is defined as:

$$\mathbf{Vod = Vd1 - Vd2, \quad Avd = Vod / Vid}$$

For common-mode excitation, both input gates are driven by the same signal. Ideally, equal drain changes cancel in the differential output, so Vod approaches zero.

$$\mathbf{CMRR = |Avd / Acm|,\ where\ Acm = Vod,cm / Vicm}$$

A larger CMRR means the circuit amplifies differential signals while rejecting noise or offsets that appear at both inputs.

2. BJT Output Characteristics

The BJT simulation uses a 2N2222 NPN transistor. The collector-emitter voltage source V1 is swept from 0 V to 15 V, while the base current source I1 is stepped from 0 µA to 100 µA in 10 µA increments. This produces a family of IC–VCE curves.

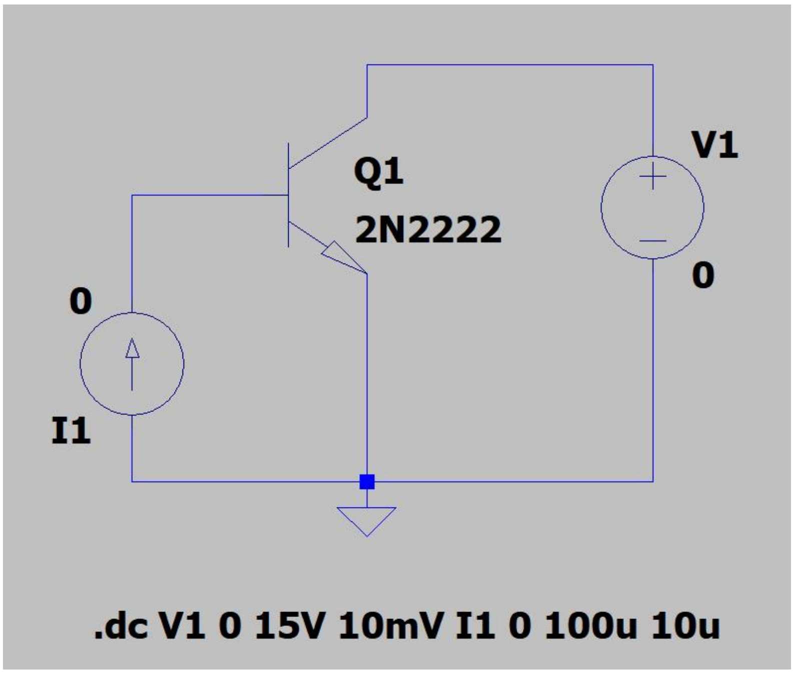

Figure 1. 2N2222 BJT DC-sweep circuit and sweep command.

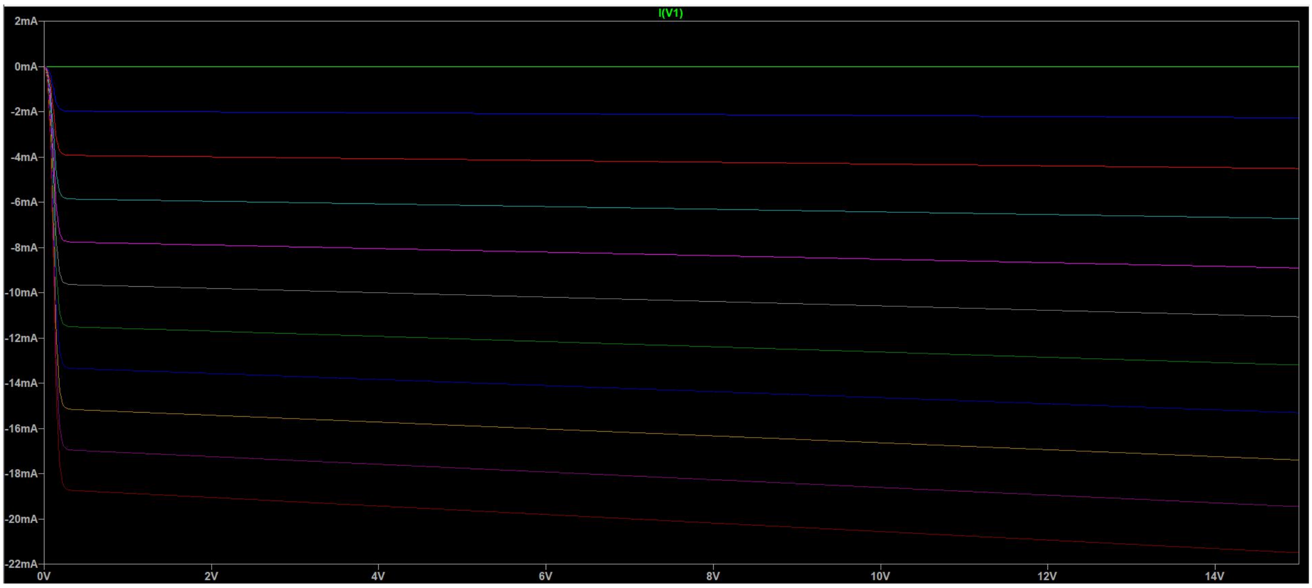

Figure 2. Current through the collector voltage source, I(V1), from the DC sweep. The negative sign is caused by LTspice source-current convention; the magnitude corresponds to collector current.

BJT observations

The characteristic curves rise sharply near VCE = 0 V and then become nearly horizontal after the saturation region. This confirms normal BJT behavior: saturation at small VCE and active operation at higher VCE. The curve spacing is approximately proportional to the stepped base current.

Base-current step	Approximate collector-current magnitude in active region	Estimated current gain
10 µA	≈ 2 mA	≈ 200
50 µA	≈ 10–11 mA	≈ 200–220
100 µA	≈ 20–21 mA	≈ 200–210

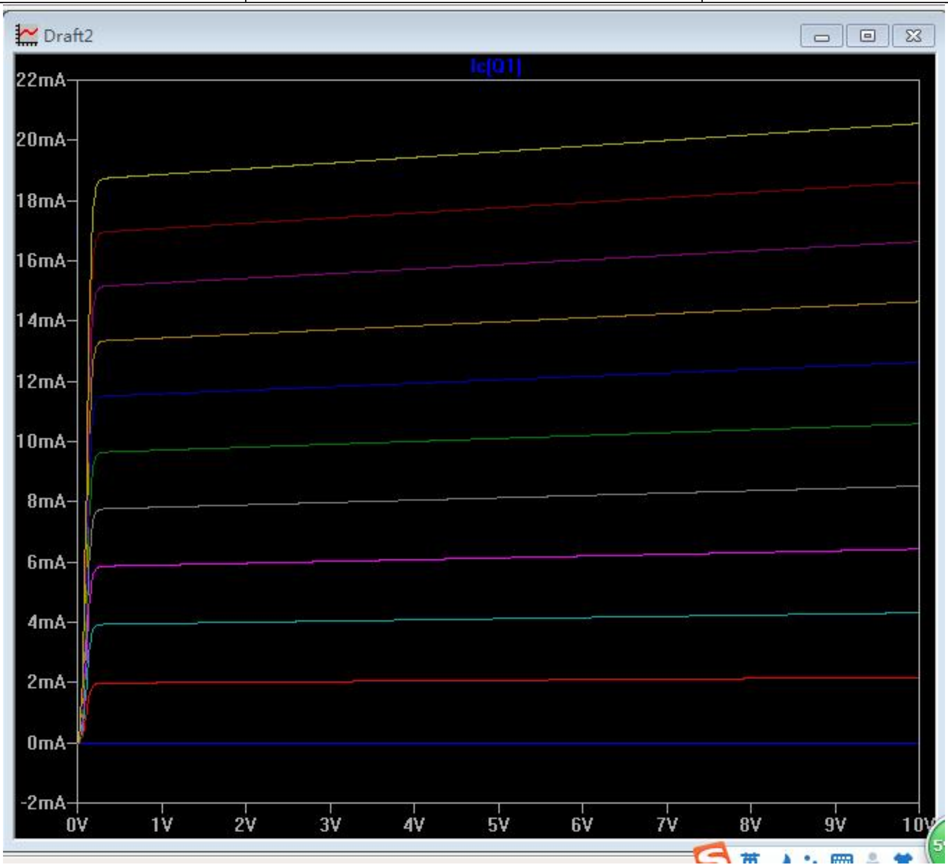

Figure 3. Positive collector-current view of the BJT output characteristics.

The upward slope of each active-region curve is small but visible, indicating finite output resistance. As base current increases, the collector-current level increases from approximately 0 mA to about 20 mA.

3. NMOS Characteristic-Curve Setup

The NMOS test circuit is arranged with one independent source controlling VGS and another controlling VDS. This is the standard arrangement for obtaining MOSFET output characteristics by sweeping drain-source voltage for several gate-source voltage values.

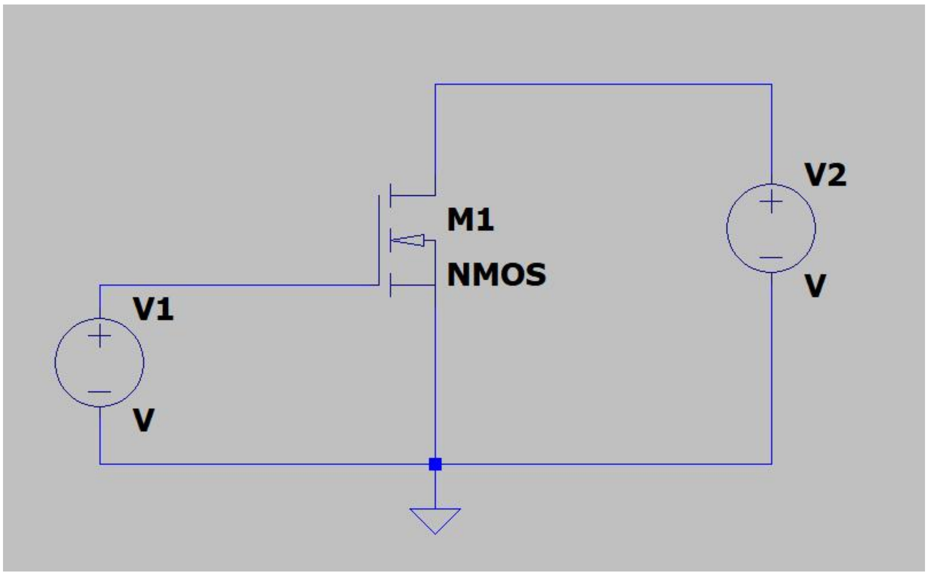

Figure 4. NMOS DC test circuit from the submitted screenshots.

A completed MOSFET output plot was not present in the submitted DOCX screenshots. Therefore, this report does not assign a numerical threshold voltage or drain-current level from the NMOS test. The expected result is a family of drain-current curves: small current when VGS is below threshold, a steep increase in the triode region, and flatter curves in saturation as VDS increases.

For a complete numerical MOSFET section, the missing LTspice plot should be generated with nested DC sweeps of VDS and VGS, then read using cursor values at selected operating points.

4. Differential Amplifier Simulation Setup

The differential amplifier uses two matched 2N7002 NMOS transistors. The drain resistors are both 10 kΩ, and the gate input resistors are also 10 kΩ. The circuit is biased from +12 V and −12 V supplies with a 2.577 mA tail current source. The transient simulation stop time is 20 ms, which covers 20 cycles of a 1 kHz input.

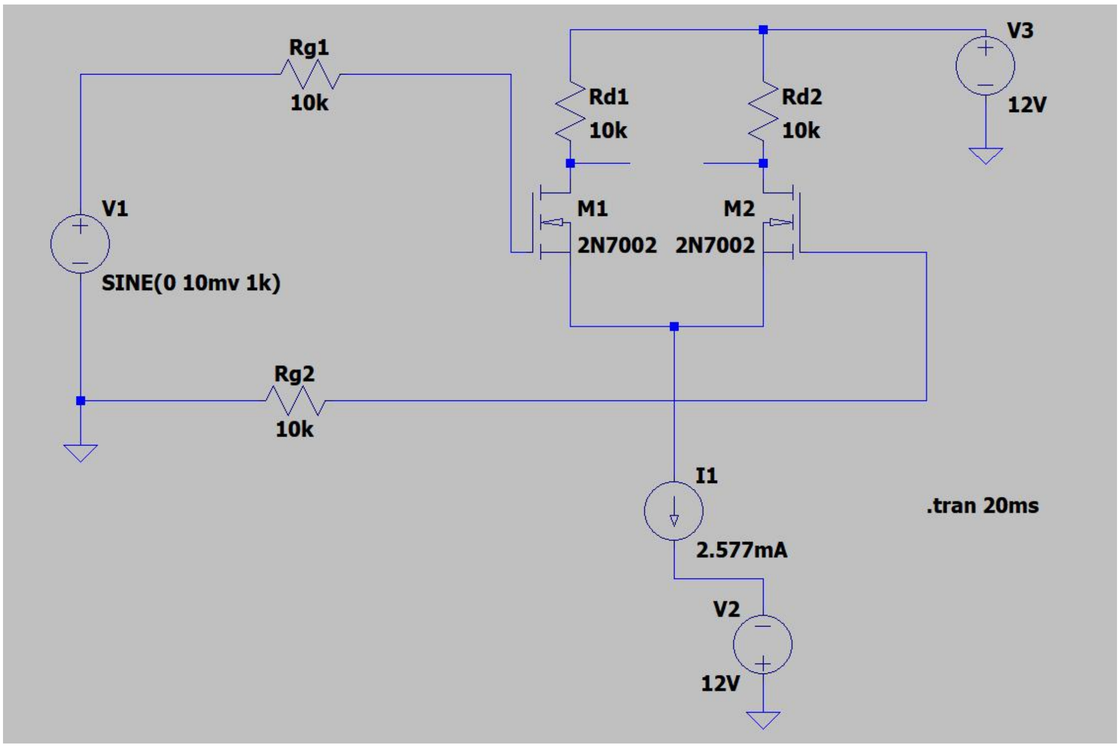

Figure 5. Differential-input amplifier circuit: SINE(0, 10 mV, 1 kHz), 20 ms transient run.

The input is applied to one side of the differential pair while the other input is referenced through the corresponding gate resistor. The two drain nodes are recorded as V(n004) and V(n005); their difference is V(n004) − V(n005).

5. Differential Gain Result

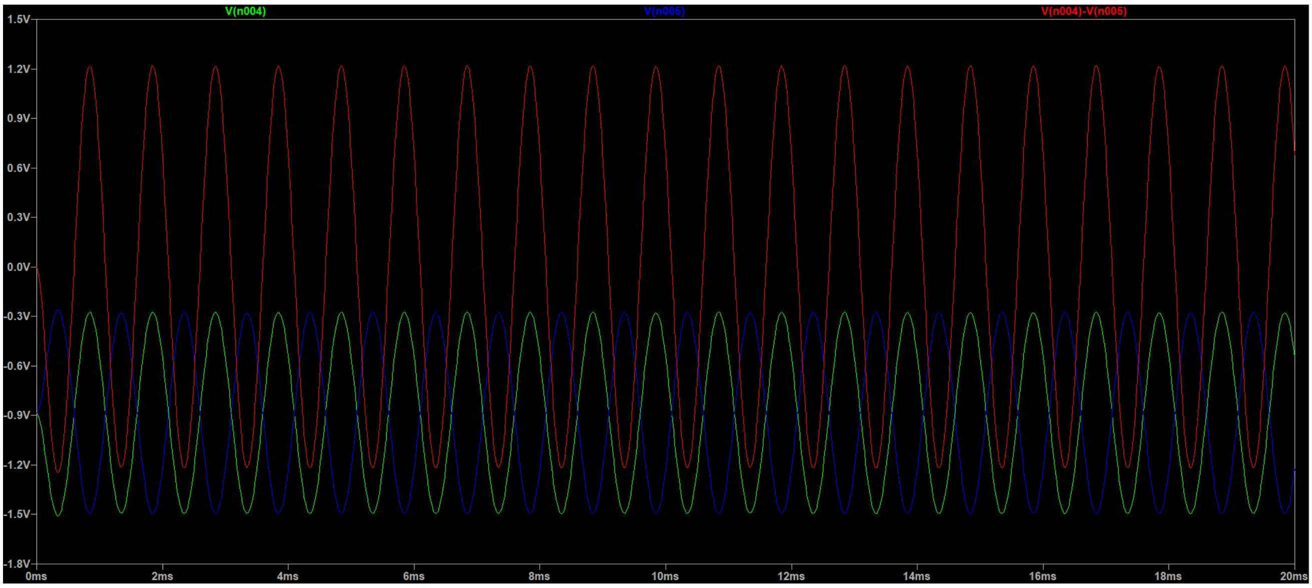

Figure 6. Differential transient waveforms. Green/blue traces are the two drain-node voltages; the red trace is V(n004) − V(n005).

From the plot, the differential-output waveform has an approximate amplitude of ±1.2 V. The applied input sine amplitude is 10 mV. Using peak amplitudes:

Avd ≈ Vod / Vid ≈ 1.2 V / 0.010 V = 120 V/V

In decibels, 20 log10(120) ≈ 40.8 dB. The two drain outputs are approximately opposite in phase, so their subtraction produces a larger differential waveform. The red trace is therefore the most direct measurement of differential gain.

The output remains periodic at 1 kHz and does not show severe clipping in the displayed range, so the selected bias point is suitable for the 10 mV input amplitude.

6. Common-Mode Rejection and Conclusion

For common-mode testing, both gates are driven by the same 1 kHz source. The submitted circuit uses a larger common-mode input amplitude of 100 mV. A well-balanced differential pair should produce nearly equal changes at both drains, making the differential output close to zero.

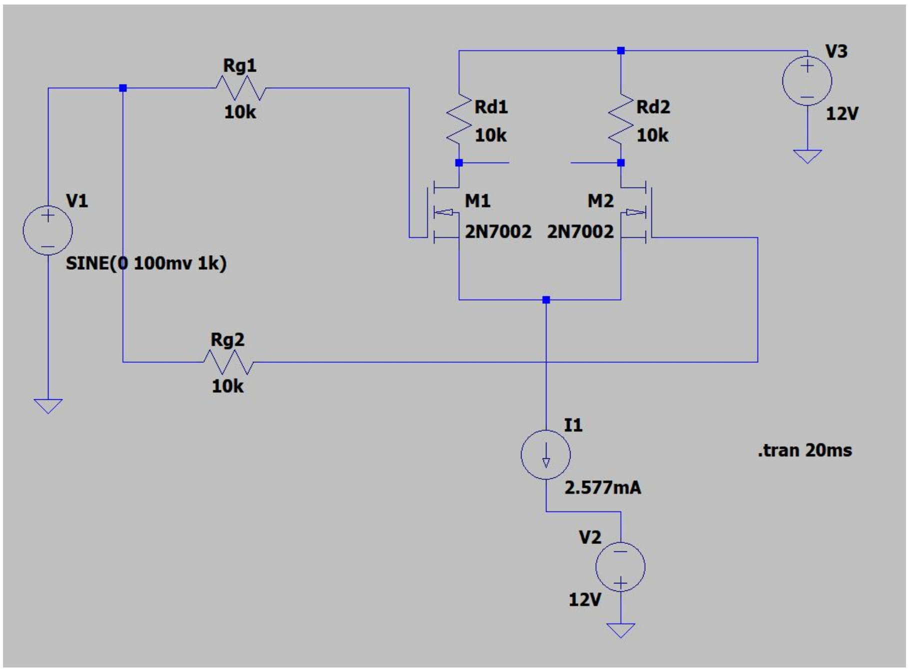

Figure 7. Common-mode input circuit: SINE(0, 100 mV, 1 kHz).

Figure 8. Common-mode transient result. The differential trace V(n004) − V(n005) is visually near zero at the plot resolution.

The graph shows that the differential output under common-mode excitation is approximately 0 mV. Because the screenshot does not include cursor data, a conservative upper bound of 1 mV is used for calculation:

$$\mathbf{A_{cm} < 0.001\ V / 0.100\ V = 0.01\ V/V; \quad CMRR > 120 / 0.01 = 1.2\times10^{4} \approx 72\ dB}$$

Conclusion. The BJT simulation confirms the expected transition from saturation to active operation and shows collector current scaling with base current. The NMOS circuit was prepared for characteristic-curve measurement, but the submitted data did not include a completed NMOS plot. The differential amplifier shows high differential gain and strong common-mode rejection. These results match the purpose of the laboratory exercise: to connect transistor device behavior with amplifier performance in LTspice.

Reference materials. Laboratory guide PDF: Fundamentals of Analogue Circuits, Experiment Part; submitted DOCX screenshots: LTspice circuits and waveform plots.